

AI Model Strategy

The objective of the AI Model Strategy is to define how artificial intelligence capabilities are selected, governed, deployed, and continuously improved within the Clinical Imaging Intelligence Platform.

A key architectural principle of this solution is that **business value is not derived from a single model**, but from the coordinated execution of multiple specialized AI models orchestrated through an agent-based workflow.

Therefore, the platform adopts a **multi-model architecture**, where each model is optimized for a specific clinical task and contributes to a broader diagnostic support process.

AI Strategy Principles

The model strategy is guided by the following principles:

Best Model for Each Task

No single model is expected to perform all imaging-related functions.

Different tasks require different architectures:

- Segmentation
- Classification
- Anomaly detection
- 3D reconstruction
- Clinical reasoning

Each capability should be implemented using the most appropriate model family.

Clinical Safety Before Automation

Models provide recommendations and analytical support.

They do not replace clinical decision-making.

All outputs remain subject to clinician review and validation.

Explainability by Design

Where possible, model outputs should provide:

- Confidence scores
- Visual evidence
- Traceable reasoning paths
- Processing metadata

This is essential for clinician trust and regulatory acceptance.

Continuous Evolution

Models should be independently replaceable and upgradeable.

The architecture must support:

- Model versioning
- Incremental improvements
- Introduction of new capabilities

without requiring platform redesign.

AI Capability Mapping

The platform is expected to support multiple AI capabilities.

Capability	AI Category	Business Purpose
Organ Segmentation	Computer Vision	Identify anatomical structures
Tissue Classification	Computer Vision	Support diagnosis
Anomaly Detection	Computer Vision	Highlight suspicious findings
3D Reconstruction	Deep Learning	Generate spatial representations
Workflow Routing	Agentic AI	Coordinate processing
Clinical Explanation	Generative AI (Future)	Support interpretation
Knowledge Retrieval	RAG (Future)	Provide clinical context

Segmentation Model Strategy

Business Objective

Automatically identify and isolate anatomical structures from medical images.

Examples:

- Heart chambers
- Coronary arteries
- Blood vessels
- Tumors
- Organs

Candidate Model Families

The most common approaches include:

- U-Net
- U-Net++
- Attention U-Net
- nnU-Net
- Vision Transformers (ViT-based segmentation)

Selection Rationale

For a production healthcare environment, architectures derived from U-Net remain highly attractive because of:

- Strong performance in medical imaging
- Extensive clinical validation
- Availability of pretrained models
- Mature research ecosystem

Success Metrics

Examples:

- Dice Similarity Coefficient (DSC)
- Intersection over Union (IoU)
- Precision
- Recall

Target:

- Dice Score > 0.90 for primary anatomical structures

Anomaly Detection Strategy

Business Objective

Identify imaging patterns that may require specialist attention.

Examples:

- Lesions
- Structural abnormalities
- Vascular irregularities
- Tumor candidates

Candidate Approaches

Supervised Detection

Suitable when labeled datasets exist.

Examples:

- CNN classifiers
- Vision Transformers
- Multi-class diagnostic models

Semi-Supervised / Unsupervised Detection

Useful for rare diseases.

Examples:

- Autoencoders
- Variational Autoencoders
- Contrastive learning approaches

Strategic Consideration

Healthcare datasets often suffer from:

- Class imbalance
- Rare pathology occurrence
- Limited annotations

For this reason, anomaly detection strategies should combine supervised and unsupervised approaches.

3D Reconstruction Strategy

Business Objective

Convert sequential 2D image slices into clinically meaningful 3D representations.

Reconstruction Approaches

Segmentation-Based Reconstruction

Process:

DICOM Slices



Segmentation

↓
Volume Reconstruction

↓
3D Mesh Generation

Advantages:

- Explainable
- Clinically interpretable
- Widely adopted

Deep Learning Reconstruction

Potential future evolution:

- Neural radiance approaches
- Implicit neural representations
- Learned volumetric reconstruction

These approaches may provide:

- Better fidelity
- Improved automation

but currently have lower clinical maturity.

Recommended PoC Approach

For the initial platform, Segmentation-driven reconstruction should be preferred due to its higher explainability and clinical acceptance.

Agentic Decision Models

Unlike imaging models, orchestration agents focus on workflow intelligence.

Their role is not image interpretation but decision-making regarding process execution.

Their value comes from:

- Context awareness
- Dynamic decision-making
- Workflow adaptation

rather than statistical image analysis.

Routing Agent

Responsibilities:

- Identify examination type
- Select appropriate processing pipeline
- Allocate computational resources

Validation Agent

Responsibilities:

- Evaluate confidence thresholds
- Detect low-quality outputs
- Trigger escalation workflows

Governance Agent

Responsibilities:

- Track model usage
 - Maintain decision lineage
 - Enforce compliance policies
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Generative AI Strategy

Although not required for the initial solution, Generative AI can provide significant future value.

Clinical Report Drafting

Potential capabilities:

- Generate preliminary findings summaries
- Structure reporting templates
- Reduce documentation effort

Clinical Question Answering

Using Retrieval-Augmented Generation (RAG):

- Retrieve relevant guidelines
- Reference historical cases
- Provide contextual explanations

Explainability Layer

LLMs may translate technical AI outputs into clinician-friendly language.

Example:

Instead of:

"Segmentation confidence: 0.91"

The system could explain:

"The identified structure is consistent with expected anatomical boundaries and matches historical segmentation patterns with high confidence."

Governance Consideration

Generative AI outputs should always remain advisory.
They must never constitute autonomous diagnostic conclusions.

Model Lifecycle Management

Healthcare AI systems require strict governance throughout the model lifecycle.

Development

Activities:

- Data preparation
- Training
- Validation
- Bias assessment

Certification

Before deployment:

- Clinical validation
- Regulatory review
- Performance benchmarking

Production

Activities:

- Monitoring
- Drift detection
- Performance auditing

Retirement

Models must be decommissioned when:

- Performance degrades
- Better alternatives emerge
- Regulatory requirements change

Evaluation Framework

The platform should evaluate models across multiple dimensions.

Dimension	Example Metrics
Accuracy	Dice, IoU, F1
Clinical Utility	Clinician acceptance rate
Safety	False negative rate
Efficiency	Processing latency
Robustness	Cross-institution performance
Explainability	Reviewability of outputs

A model should not be considered successful solely because of technical accuracy. Clinical usefulness and operational reliability are equally important.

Strategic AI Evolution Roadmap

The architecture is intentionally designed to evolve over time.

- **Phase 1:** AI-assisted segmentation and reconstruction.
- **Phase 2:** Advanced anomaly detection.
- **Phase 3:** Clinical knowledge retrieval and RAG.
- **Phase 4:** Multi-agent clinical reasoning support.
- **Phase 5:** Personalized predictive healthcare models.

AI Strategy Statement

The proposed platform adopts a multi-model, agent-orchestrated AI strategy that combines specialized computer vision models, workflow intelligence agents, and future generative AI capabilities into a clinically governed ecosystem.

Rather than pursuing a single monolithic AI solution, the architecture emphasizes modularity, explainability, and continuous evolution, ensuring that AI capabilities can adapt alongside medical knowledge, regulatory requirements, and organizational needs while maintaining clinical trust and patient safety.